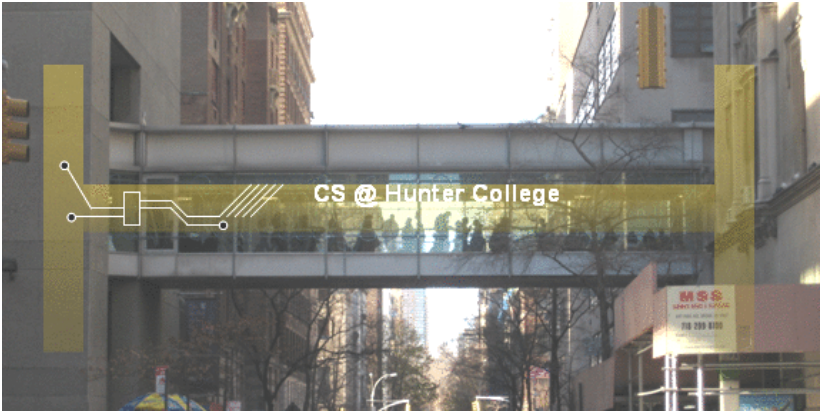
CSci 127: Introduction to Computer Science



hunter.cuny.edu/csci

Outline

- Syllabus & Course Overview
- Introduction to Python
- Turtles & Definite Loops (`for`-loops)
- Algorithms
- Wrap Up

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  We will automatically drop and/or replace lowest grades (see syllabus). If you don't miss any, we will still drop and replace lowest grades.

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  How to Think Like a Computer Scientist.

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Course Description

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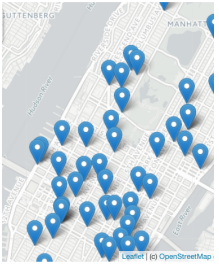
Catalog Description: 3 hours, 3 credits: This course presents an overview of computer science (CS) with an emphasis on ***problem-solving and computational thinking through 'coding'****: computer programming for beginners.*
This course is pre-requisite to several introductory core courses in the CS Major. The course is also required for the CS minor. MATH 12500 or higher is strongly recommended as a co-req for intended Majors.

Syllabus: Topics



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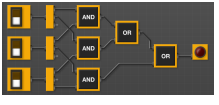

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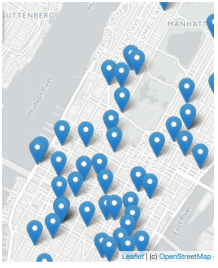

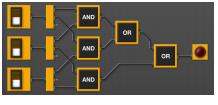

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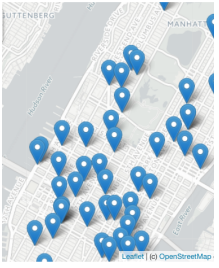

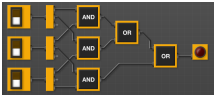

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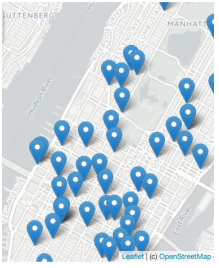

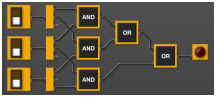

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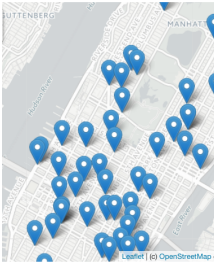

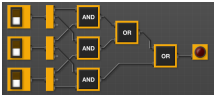

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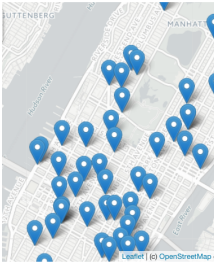

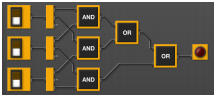

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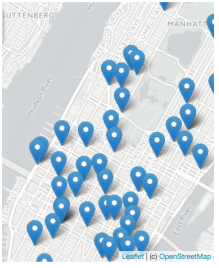

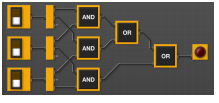

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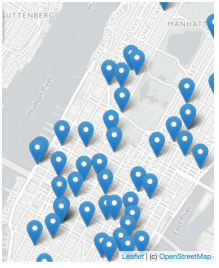

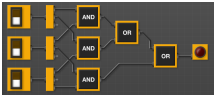

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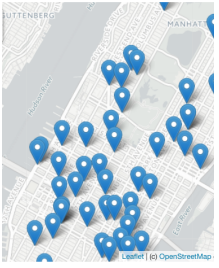

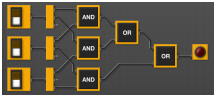

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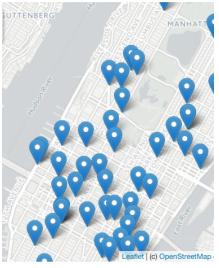

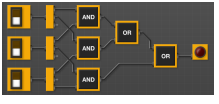

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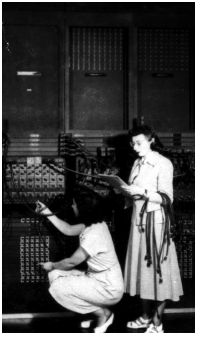

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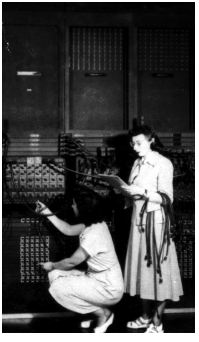

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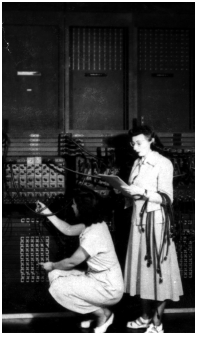

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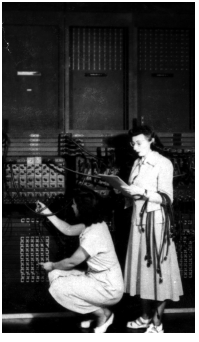

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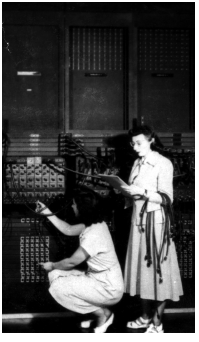

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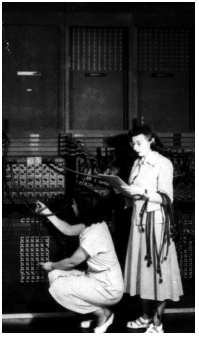

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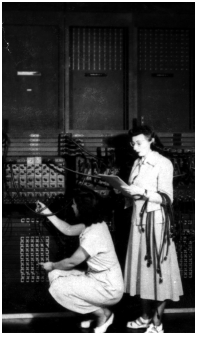

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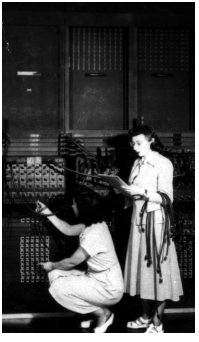

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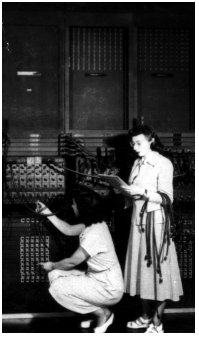

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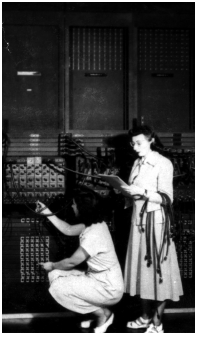

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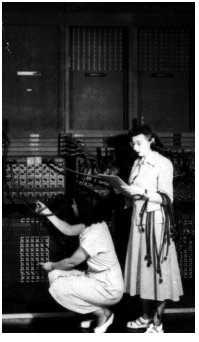

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  Yes. To demonstrate mastery, you must pass the final exam.
  We will end most lectures with past final exam questions and review.

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  - *Work through the labs.*

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- The first lab goes into step-by-step details of getting Python running.
- We'll look at the design and basic structure (no worries if you haven't tried it yet).

First Program: `Hello, World!`

Demo in `pythonTutor`

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```
#Name: Thomas Hunter
#Date: May 28, 2026
#This program prints: Hello, World!

print("Hello, World!")
```

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← *Prints*

- Output to the screen is: `Hello, World!`

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← *Prints*

- Output to the screen is: `Hello, World!`
- We know that `Hello, World!` is a **string** (a sequence of characters) because it is surrounded by quotes
- Can replace `Hello, World!` with another string to be printed.

Variations on `Hello, World!`

```
#Name: L-M Miranda
#Date: Hunter College HS `98
#This program prints intro lyrics

print('Get your education,')
```

Spring, 2018, Assembly Hall

Variations on `Hello, World!`

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#Name: L-M Miranda
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print('Get your education,')
print("don't forget from whence you came, and")
print("The world's gonna know your name.")
```

- Each print statement writes its output on a new line.
- Results in three lines of output.
- Can use single or double quotes, just need to match.

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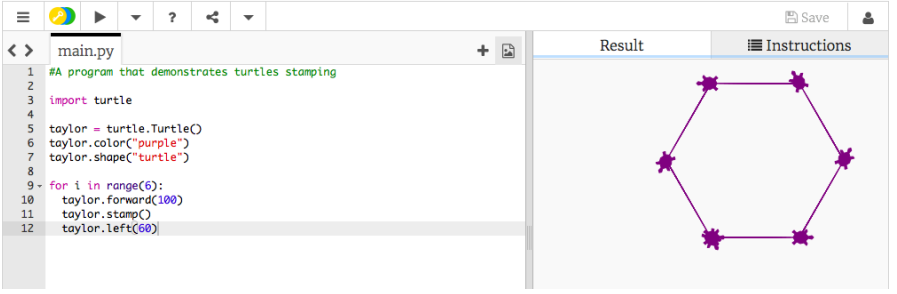
Turtles & Definite Loops



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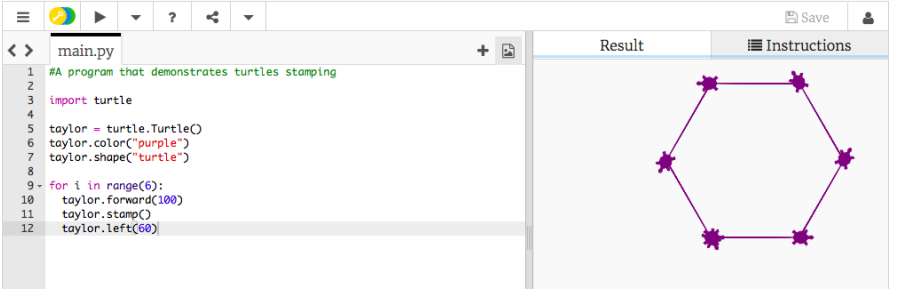

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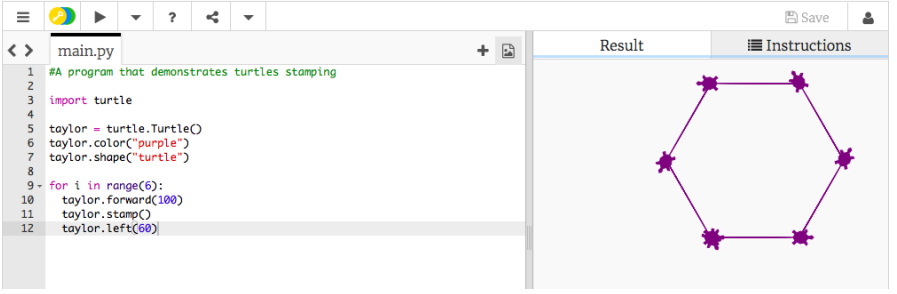

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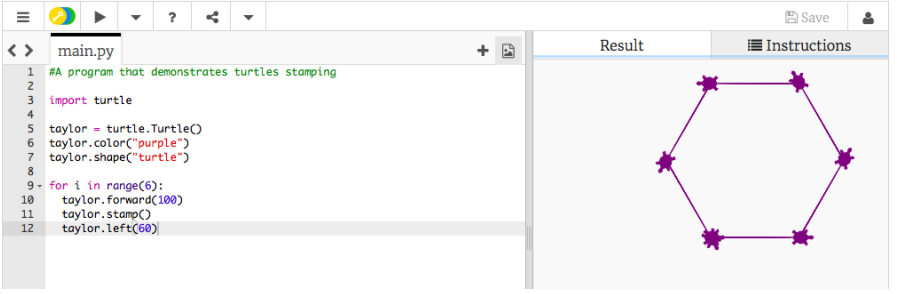
- Repeats 6 times:

Turtles & Definite Loops



- Creates a turtle **variable**, called `taylor`.
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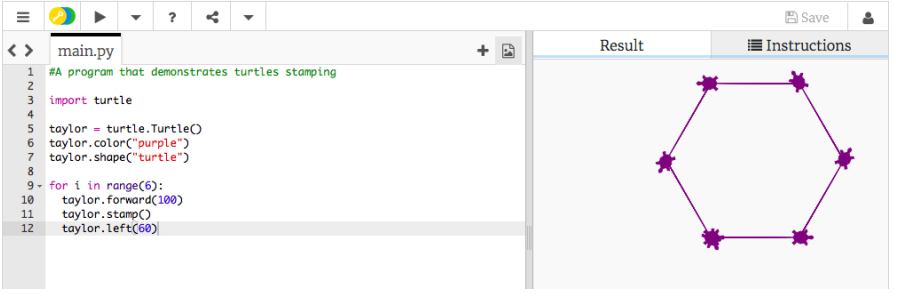
- Repeats 6 times:
  - Move forward; stamp; and turn left 60 degrees.

Turtles & Definite Loops



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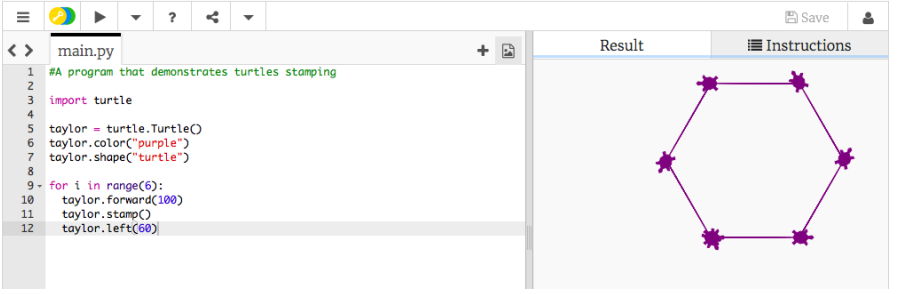

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- Repeats any instructions **indented** in the "loop block"
- This is a **definite** loop because it repeats a fixed number of times

Group Work

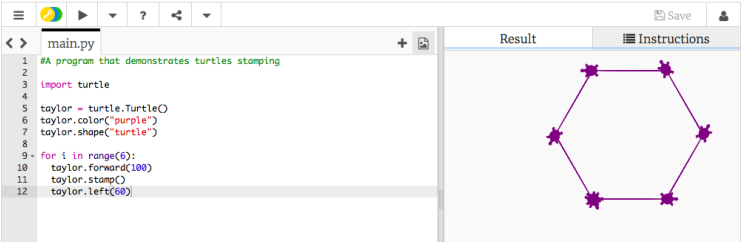
Working in pairs or triples:

1. Write a program that will draw a 10-sided polygon.
2. Write a program that will repeat the line:

   ```
   I'm lookin' for a mind at work!
   ```

   three times.

Decagon Program



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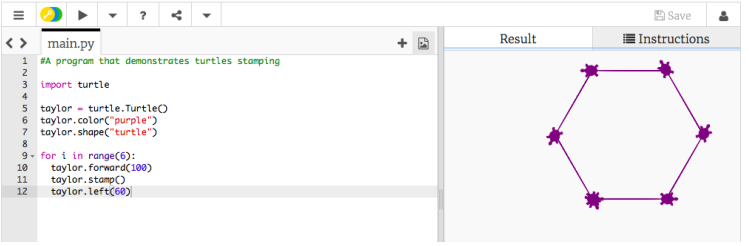

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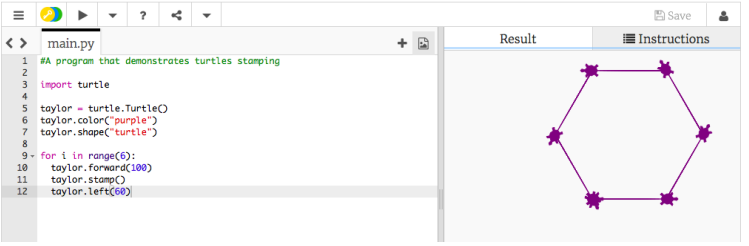

- Start with the hexagon program.
- Has 10 sides (instead of 6), so change the `range(6)` to `range(10)`.
- Makes 10 turns (instead of 6),
  so change the `taylor.left(60)` to `taylor.left(360/10)`.

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  ```

- Instead of turtle commands, repeating a print statement.
- Completed program:

  ```
  # Your name here!
  for i in range(3):
      print("I'm lookin' for a mind at work!")
  ```

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What is an Algorithm?

From our textbook:

- An algorithm is a process or sequence of steps to be followed to solve a problem.

What is an Algorithm?

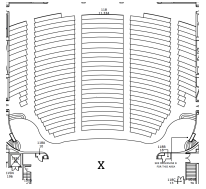
From our textbook:

- An algorithm is a process or sequence of steps to be followed to solve a problem.
- Programming is a skill that allows a computer scientist to take an algorithm and represent it in a notation (a program) that can be executed by a computer.

Outline

- Syllabus & Course Overview
- Introduction to Python
- Turtles & Definite Loops (`for`-loops)
- Algorithms
- Wrap Up

Recap



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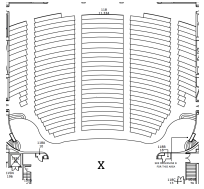

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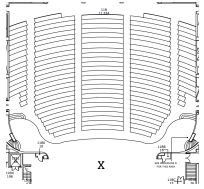

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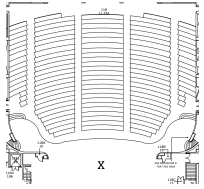

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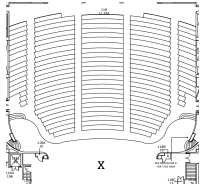

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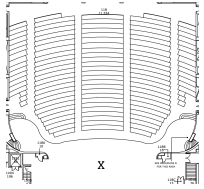

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  - strings, or sequences of characters,
  - `print()` statements,
  - `for`-loops with `range()` statements, &
  - variables containing turtles.

Remainder of lecture time

- Make sure you have access to the Gradescope before leaving.
- Refer to lab 1 for setting up python and writing first programs.
- Use this time to work on the first 4 assignments (due Friday + Monday).
- (Assignment 5 due date will be pushed back to next Tuesday).
- It is good practice to work on the assignments in advance! Gradescope can and will cause unexpected issues. Please use this time to ask questions when stuck.